

Day 1 — Finger-Trick Prelude

Module: Multiplication Near a Power of 10 — Nikhilam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: practice the below-10 finger method (e.g., 7×8 using deficits).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Nikhilam* (IAST: Nikhilam; lit. “the Nikhilam sūtra (also used in Module 9 for subtraction)”)

Primary source (today’s reading)

From *Bharati Krishna Tirthaji* — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Recall Module 9.** Nikhilam = “all from 9, last from 10” — the complement to a base. Same sūtra, NEW application: multiplication.
2. **The below-10 finger demo.** Each hand has 5 fingers; together 10. To multiply 7×8 :
 - Curl $7-5=2$ fingers on left hand.
 - Curl $8-5=3$ fingers on right hand.
 - Curled fingers in total: 5 → that’s the tens place: 50.
 - Uncurled fingers (left $\times$ right): $3 \times 2 = 6$ → that’s the units: 6.
 - Answer: 56. ✓
3. **Why this works.** Algebraically: $(10 - a)(10 - b) = 100 - 10(a+b) + ab = 10(10-(a+b)) + ab$. The “curled fingers” count is $(a+b)-5$ — wait, let me re-derive. Actually: curled = $(10-a-5) + (10-b-5)$ wait... let me restate cleanly. For 7×8 : $7 = 5+2$ (uncurled fingers

on left), $8 = 5+3$ (right). The formula: tens digit = uncurled-left + uncurled-right – 5; units = uncurled-left $\times$ uncurled-right. Let’s just use the Nikhilam algebra in next lesson.

4. **What we’ll learn this module.** A general method: when factors are CLOSE to a base (10, 100, 1000), Nikhilam-multiplication is much faster than long multiplication.
5. **Speed test.** Compute 8×9 mentally. Compare your method with traditional times table. Most students will use memorisation — the next lessons show why Nikhilam is faster for LARGER near-base products.

Activity (15 min)

A scaled version of *Method Speed Race* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Finger-Trick Prelude.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).